Unit: 6 Unit 4: Absolute Value **Name:** ________________
Topic: Absolute Value Inequalities and Their Applications **Date:** __________

Homework (Habanero)

Q1. Solve the inequality $|x-2| leq 3$.
(A) $-1 leq x leq 5$
(B) $-4 leq x leq 1$
(C) $0 leq x leq 4$
(D) $1 leq x leq 3$
(E) $2 leq x leq 6$

Q2. The dosage of a certain medicine is $|d-10| leq 2$. What is the range of possible dosages?
(A) $8 leq d leq 12$
(B) $6 leq d leq 10$
(C) $10 leq d leq 14$
(D) $12 leq d leq 16$
(E) $8 leq d leq 10$

Q3. Graph the solution set of the inequality $|x| > 2$ on a number line.
(A) $(-infty, -2) cup (2, infty)$
(B) $[-2, 2]$
(C) $(-infty, 2]$
(D) $[2, infty)$
(E) $[-2, 2)$

Q4. A certain bacteria grows at a rate of $|r-5| leq 1$. What is the range of possible growth rates?
(A) $4 leq r leq 6$
(B) $3 leq r leq 5$
(C) $5 leq r leq 7$
(D) $6 leq r leq 8$
(E) $4 leq r leq 8$

Q5. The error in a medical measurement is $|e-0.5| leq 0.2$. What is the range of possible errors?
(A) $0.3 leq e leq 0.7$
(B) $0.2 leq e leq 0.5$
(C) $0.5 leq e leq 0.8$
(D) $0.1 leq e leq 0.4$
(E) $0.3 leq e leq 0.6$

Q6. Solve the inequality $|x+1| < 4$.
(A) $-5 < x < 3$
(B) $-3 < x < 1$
(C) $-1 < x < 5$
(D) $-4 < x < 2$
(E) $-2 < x < 4$

Q7. The concentration of a certain substance is $|c-2| geq 1$. What is the range of possible concentrations?
(A) $c leq 1$ or $c geq 3$
(B) $1 leq c leq 3$
(C) $c geq 2$
(D) $c leq 2$
(E) $c geq 1$

Q8. Graph the solution set of the inequality $|x-1| > 2$ on a number line.
(A) $(-infty, -1) cup (3, infty)$
(B) $[-1, 3]$
(C) $(-infty, 1)$
(D) $[3, infty)$
(E) $(-infty, -1] cup [3, infty)$

Open Ended Questions (FRQ)

Q9. A patient's temperature is $|t-98| leq 1$. Find the range of possible temperatures and express it in interval notation.

Q10. The growth rate of a certain tumor is $|r-2| geq 1$. Find the range of possible growth rates and graph it on a number line.

Chef's Tips

- Read each question carefully before answering.
- Use a number line to help visualize the solution set.
- Check your work and make sure your answer is reasonable.